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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

EC4331-1 – LOW POWER VLSI DESIGN
(For ECE)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following best explains why low-power design is essential for portable devices?
A) To reduce chip cost
B) To minimize fan noise
C) To extend battery life
D) To comply with Moore's Law
2. Which factor directly increases packaging and cooling costs in high-performance chips?
A) Higher transistor density
B) Higher power dissipation
C) Lower supply voltages
D) Miniaturization of circuits
3. Leakage power problems mainly occur in:
A) High-frequency circuits with heavy switching activity
B) Very low-frequency circuits or circuits in sleep modes
C) Circuits with high transistor density only
D) Circuits with high packaging costs
4. Which of the following is not a method for reducing static current?
A) Transistor sizing
B) Layout techniques
C) Careful circuit design
D) Increasing supply voltage
5. One effective way to reduce static current in unused circuit modules is to:
A) Reduce transistor size
B) Turn them OFF
C) Reverse clock polarity
D) Increase switching frequency
6. The main advantage of probabilistic power analysis over simulation-based methods is:
A) Higher accuracy
B) Lower power consumption
C) Higher computation efficiency
D) Better transistor-level modeling
7. In probabilistic power analysis, a logic signal is modeled as:
A) Deterministic time-domain waveform
B) Random zero-one process with statistical properties
C) Continuous analog voltage waveform
D) Ideal square wave
8. The drawback of probabilistic power analysis compared to simulation is:
A) High complexity
B) Increased computation time
C) Loss of accuracy
D) Larger circuit area
9. Probabilistic analysis is mainly applied at which abstraction level:
A) Transistor level
B) Device level
C) Gate level and above
D) Physical layout level
10. The switching frequency f of a digital signal is defined as:
A) Number of logic highs per second
B) Number of transitions per second
C) Half the number of transitions per unit time
D) Product of voltage and capacitance
11. Reversing the signal polarity in certain cases helps to:
A) Reduce propagation delay
B) Increase dynamic activity
C) Minimize the probability of static current flow
D) Increase supply rail noise immunity
12. The amount of static current in a circuit depends on:
A) Transistor channel length
B) Logic state of the output
C) Packaging technology
D) Battery type used

13. One effective way to reduce static current in unused circuit modules is to:
 A) Reduce transistor size B) Turn them OFF
 C) Reverse clock polarity D) Increase switching frequency
14. Which of the following is not a method for reducing static current?
 A) Transistor sizing B) Layout techniques
 C) Careful circuit design D) Increasing supply voltage
15. Leakage power problems mainly occur in:
 A) High-frequency circuits with heavy switching activity B) Very low-frequency circuits or circuits in sleep modes
 C) Circuits with high packaging costs D) Circuits with high transistor density only
16. Static probability p of a signal represents:
 A) The probability that the signal is logic 1 B) The switching frequency
 C) The rise and fall delay ratio D) The average power of the signal
17. The relationship between static probability p and expected frequency f for a memoryless random signal is:
 A) $f = p^2$ B) $f = 2p(1-p)$
 C) $f = p(1-p)$ D) $f = (1-p)^2$
18. For a memoryless random signal, the maximum frequency occurs when:
 A) $p = 0.25$ B) $p = 0.75$
 C) $p = 0.5$ D) $p = 1$
19. The concept of lag-one signal model introduces:
 A) One more degree of freedom using conditional probabilities B) Infinite memory of previous states
 C) Transistor-level modeling D) Static power computation
20. In lag-one model, the transition probability T is given by:
 A) $T = 2p(1-p)$ B) $T = p10 + p01$
 C) $T = p11 + p00$ D) $T = p^2 + (1-p)^2$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain in detail the methodology and limitations of SPICE-based power analysis in digital VLSI design.	8	L*2	1	1
	b) Discuss the advantages, disadvantages, and accuracy trade-offs among different levels of simulation abstraction.	8	L2	1	2
2.	a) Describe the procedure for gate-level power estimation and explain how capacitive, internal, and static components are obtained.	8	L2	1	1
	b) Elaborate on the Dual Bit-Type Model and its significance in power estimation of DSP systems with correlated data.	8	L2	2	2
3.	a) Explain the concept and importance of Monte Carlo power simulation and describe how confidence level and error tolerance affect the stopping criteria.	8	L2	1	1
	b) The standard deviation of the power samples measured from a circuit has been observed to have $\pm 20\%$ fluctuation from the mean. How many samples are required so that we are 95% confidence that the error of sample mean is within $\pm 5\%$? Given Data : $(1 - \alpha) = 0.95$, we have $Z_{\alpha/2} = 1.96$	8	L3	2	2
Unit – II					
4.	a) Explain the concept and importance of Boolean difference (dy/dxi) in the propagation of transition density. How is this applied in calculating output activity of a logic circuit?	8	L3	4	2
	b) Explain and plot the graph of inverter chain delay and power dissipation.	8	L2	3	1

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| 5. a) | Explain the Sizing an Inverter Chain with a neat diagram and equations. | 8 | L2 | 4 | 2 |
| b) | Compare transition density and entropy-based approaches for power estimation. Discuss the level of abstraction, computational cost, and accuracy of both methods. | 8 | L2 | 3 | 1 |
| 6. a) | With a neat diagram, explain the entropy analysis of a sequential circuit. | 8 | L2 | 4 | 1 |
| b) | Compute the transition density and static probability of $y=ab + c$ given $P(a)=0.2$, $P(b)=0.3$, $P(c)=0.4$, $D(a)=1$, $D(b)=2$, $D(c)=3$ | 8 | L3 | 4 | 2 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Describe the Power Dissipation in Clock Distribution and write an equation for the dynamic power dissipated by switching the clock. | 8 | L2 | 5 | 1 |
| b) | Describe the derivation of Tolerable Skew. | 8 | L2 | 5 | 1 |
| 8. a) | Explain the adaptive performance management with a neat diagram. | 8 | L2 | 5 | 1 |
| b) | With a neat filter architecture explain the adaptive filter. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
